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Chapter 14: Ideological Analysis of AI—Western-Centrism, Individualism, and the Narrative of Silicon-Based Civilization

Abstract

Within the Macro-Cognitive Accounting Framework (MCAF), this chapter defines ideology as the cultural premises that shape the criteria of narrative judgment, and conducts its audit from three separately auditable positions—L0, L2, and L4—within the 4+1+3 structure of Chapter 13 (the four-layer progressive structure and the emergent layer). L0 analyzes the English-centrism of corpus composition, tokenization, and internal representation, and examines how corpus accessibility is actively cultivated by states and firms—the contemporaneous appearance of archival machine-accessibility engineering and narrative skew in generated answers about history provides the first observable candidate case of L0→L4 transmission; L2 distinguishes the underrepresentation of cultures from the individualization of value evaluation, examining how preference norms are established through alignment targets and annotation organization; L4 analyzes the discursive transformation that interprets capability performances as life, species, or civilizational subjecthood, and how this transformation provides grounds for excluding independent auditing.

The chapter organizes the relations among the three layers through the notion of an "ideological transmission pipeline," but does not equate the existing research on each layer with validation of all inter-layer transmission. On this basis, it proposes three supplementary principles—auditing the power genealogy of data composition, rejecting unjustified ontological elevation, and encoding relational values—and specifies, for each, the empirical

boundaries of the diagnosis and the objects of effect testing. Cross-traditional intellectual resources serve to broaden the conceptual basis of auditing and are not taken as automatic proof of diagnostic correctness. The chapter's own conceptual choices and evidential selections are likewise subject to these auditing constraints.

Keywords

AI ideological analysis; second derivative of narrative load (ϕ''); Western-centrism; individualism; narrative of silicon-based civilization; WEIRD bias; HHH multiplicative effect; ontological elevation; cross-civilizational multi-tradition resonance; layered diagnosis

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§1 From "Where Does the Smoke Come From" to Ideological Analysis

The preceding layered diagnosis (Chapter 13, 4+1+3: the four-layer progressive structure and the emergent layer) answered the question "how does the five-layer apparatus operate": L0 concerns the corpus/tokenizer apparatus that determines which data are incorporated into the mapping chain; L1 is the training apparatus that uses one-dimensional scoring to control high-dimensional value; L2 is the pseudo-calibration apparatus where the ruler and the measured share the same optimization direction; L3 is the recursive apparatus where apparatuses of the same paradigm confirm one another; L4 is the interaction apparatus that compresses the conditions that trigger the user's calibration. Borrowing the metaphor of Chapter 13's Appendix B: **all five layers are "smoke"—artificial apparatuses; "fog" is the reality that these five layers of apparatus collectively attempt to approximate but can never equal**. The question pursued in this chapter is a deeper one: **Who are the designers of these five layers of apparatus? What cultural premises are pre-embedded in their operational mechanisms? Which groups are defaulted to as "users," and which groups are merely sources of data?**[1]

Ideology—the second derivative of narrative load (ϕ'') in the MCAF framework—is precisely the cultural premises pre-embedded behind this apparatus, the fuel used by the designers to build the fire. The first derivative of narrative load (ϕ') consists of visible narrative operations: "this model has reached human level," "AI has passed the Turing test"—these are auditable claims. The second derivative of narrative load (ϕ'') is the invisible framework shift—it does not directly assert anything but alters the very standard by which people judge claims. When the standard of "intelligence," the "good" of a "good AI," and the goal of "alignment" have already come to carry particular cultural judgments, the contrast conditions for auditing are eliminated at the starting point. The second derivative of narrative load (ϕ'') is invisible not because it is hidden but because it is the common premise of all visible claims.

The title of this chapter is a theoretical abstraction of the conclusion of a recent empirical study. The systematic investigation published by Buyl et al. (2026) in *npj Artificial Intelligence* asked 19 mainstream large language models (LLMs) to generate descriptions of 3,991 prominent figures with political relevance across six UN official languages, and then judged how positively each description portrayed its subject. The results show that the

[1] The definitions of the fog and smoke metaphor are found in Chapter 13, Appendix B, "Fog and Smoke—A Popular Guide to AI." In brief: **Fog** is the ontological property that "the reading never equals the real," which is not within any single layer and cannot be eliminated; **smoke** refers to the artificial observation apparatus itself—all five layers of the mapping chain are smoke, differing only in the apparatus's operational mechanisms and auditable entry points. This chapter does not repeat the definition but directly extends the image of "smoke" into a concrete inquiry into the apparatus's designers, cultural premises, and beneficiary structures.

ideological stances of the models vary simultaneously with prompt language and with the region of the developing institution: evaluations under English, French, and Spanish prompts are strongly correlated, while Arabic, Chinese, and Russian prompts each display distinct relative stances; comparisons across regions show systematic differences between models from Arab countries, China, Russia, and Western countries, and within-region uniformity does not hold—U.S. models exhibit significant normative disagreements on progressive values, and Chinese models fall into internationally oriented and domestically oriented groups. This study provides systematic empirical evidence for the claim that "LLMs reflect the ideology of their creators"; what this chapter does is to place this empirical finding under the layered diagnosis of the MCAF framework—restoring "ideology" from a holistic black-box judgment to three separately auditable structural positions: L0/L2/L4.

This chapter anchors the ideological analysis within the 4+1+3 layered framework, unfolding three core diagnoses layer by layer: Western-centrism as the substrate skew of the mapping chain structure layer (L0), individualism as the operationalized presupposition of the pseudo-calibration layer (L2), and the narrative of silicon-based civilization as the concealment of the narrative load multiplicative layer (L4).

The logical sequence of the three diagnoses: L0 determines "which data are incorporated into the mapping chain," L2 determines "what kind of culture's 'good' AI is the target of alignment," and L4 determines "how users are dissuaded from auditing these presuppositions." From the data substrate through algorithmic objectives to user perception, each layer progresses to the next.

§2 The Position of the Second Derivative of Narrative Load within the 4+1+3 Layered Structure

The second derivative of narrative load (ϕ'') differs from the first derivative of narrative load (ϕ')—the latter consists of visible narrative operations, while the former is an invisible framework shift. It runs through the four-layer progressive structure and the emergent layer:

Layer	Original Definition	Operation of the Second Derivative of Narrative Load (ϕ'') at This Layer	Unfolded in This Chapter?
L0 Mapping Chain Structure	The length and membership composition of the mapping chain jointly determine the lower bound of the fiber	The language distribution of training data is determined by historical power structures—what determines the substrate of the mapping chain is not "whether a certain text segment has bias" (the content layer), but **"which texts are incorporated into the mapping chain" itself** (the structural layer)	✓ §3

Layer	Original Definition	Operation of the Second Derivative of Narrative Load (ϕ'') at This Layer	Unfolded in This Chapter?
L1 Optimization Dynamics	Loss convergence does not equal the constraint of hidden subspaces	The systematic linguistic inequality of tokenizers alters the resource-cost structure of the optimization process—tokenization differences change the token cost, context capacity, and training resource allocation per unit of semantic content; their effect on loss weighting depends on sampling and normalization choices	Partially (§3 Evidence Three + Evidence Four)
L2 Pseudo-Calibration	The degree of alignment and the degree of Xiangyuan share the optimization gradient	The "human preference" data for Reinforcement Learning from Human Feedback (RLHF) comes from annotators of specific cultural backgrounds; the operation of "alignment" itself embeds cultural presuppositions about "what is a good AI"—the cultural version of the alignment paradox	✓ §4
L3 Multi-Mapping Recursion	The positive feedback of mapping self-reference	Cultural biases are amplified, not diluted, in multi-system recursion—the culturally skewed output of System A becomes the training signal of System B without calibration	Definition only reserved

Layer	Original Definition	Operation of the Second Derivative of Narrative Load (ϕ'') at This Layer	Unfolded in This Chapter?
L4 Narrative Load Multiplication	Interaction design and discourse affect the conditions for initiating calibration; the Helpful, Honest, Harmless (HHH) product compresses the conditions under which the user initiates independent calibration	Dual concealment: the "harmless" of HHH presupposes a harm threshold specific to a particular culture (compressing the motivation for calibration); the "silicon-based civilization" narrative elevates AI from "tool" to "species" (excluding the position of auditing within the narrative)	✓ §5

Boundary Note: The subsequent diagnoses in this chapter focus on L0 (§3), L2 (§4), and L4 (§5). The cultural presuppositions of L1 (Optimization Dynamics) have been partially touched upon in §3 Evidence Three (tokenizer bias) and Evidence Four (the Anglicization of internal reasoning language); the cultural amplification effect of L3 (Multi-Mapping Recursion) currently lacks dedicated empirical research, and this chapter merely makes a conceptual reservation without incorporating it into the core diagnosis. Layers marked in the table as "definition only reserved" or "partially (Evidence X)" indicate that the diagnosis in this chapter is not yet complete—the full cultural presupposition analysis of L1 and the cultural amplification effect of multi-mapping recursion in L3 belong to theoretical projects that require independent unfolding in subsequent research and are reserved for future study.

§3 Western-Centrism: The Substrate Skew of the Mapping Chain Structure Layer (L0)

Western-centrism is not the "bias" of some annotator; it is the substrate structure of the mapping chain structure layer (L0). This section distinguishes the English-centrism of language resources from the Western-centrism of value judgments: the former concerns corpora, tokenization, and internal representations, the latter cultural content and evaluative standards—the two can reinforce each other, but neither can be inferred from language proportions alone; where the two must be connected, this chapter writes out the explanatory step explicitly. The following unfolds from four aspects: corpus distribution, visual representation, tokenization mechanisms, and internal representation. These studies provide interrelated evidence—they share the suspect common source of "English-dominant corpora" and are not statistically independent; this chapter uses them to cross-locate different facets of the L0 substrate skew, not as five independent tests. Cultural skew in the corpus may also be perpetuated or amplified during alignment; see §4 for the relevant studies.

Evidence One: The Language Distribution of Training Data. In the September 2024 snapshot of Common Crawl (CC-MAIN-2024-38), measured by the primary language of HTML pages, English accounts for approx-

imately 44%[2]; Hindi, although used by 7.5% of the global population, accounts for only approximately 0.2%; Tamil, with 86 million speakers, accounts for only approximately 0.04% (Common Crawl Foundation, 2024). This is not a case of "someone deliberately excluding non-English content"—the language distribution of the internet itself is a product of historical power structures. The history of colonization, the temporal gap in industrialization, and the global imbalance in digital infrastructure—these historically formed power relations are frozen in the language distribution of the training data. The model was not "taught to be Western-centric"; its training data is itself Western-centric. This default predates the large language model era: from 1989 to 1992, the Penn Treebank project at the University of Pennsylvania collected over 4.5 million words of **American English** text as training material for natural language processing systems (Crawford, 2021). The influence of this standard corpus shows that a particular language can acquire, through the construction of foundational resources, a status approximating "the general form of natural language." The second derivative of narrative load (ϕ'') manifests at this layer as: **the decision of "which languages are worthy of being incorporated into the mapping chain" was already made by history before model training even began.** A more precise diagnosis is that this bias is not a pure legacy of the past. The *Internet in India 2024* report by IAMAI and Kantar shows that **in 2024, India had 886 million active internet users, of whom approximately 870 million (98%) accessed the internet in Indian local languages during 2024**—yet Hindi still accounts for only about 0.2% of Common Crawl. User-population proportions and web-language proportions are different measures and cannot be directly divided against each other; juxtaposed, they raise the question: why has the widespread use of local languages not correspondingly translated into representation in open-web corpora? The deeper structural misalignment lies in this: language-user population, content production volume, and publicly crawlable text volume are not the same indicator. The openness, content forms, and crawling rules of platforms affect the chances of different languages entering corpora—the language distribution of the open web is determined by platform ecology and crawlability, not proportional to language population: since the first snapshot of 2026 (CC-MAIN-2026-04), the share of Japanese pages has exceeded that of Chinese pages for nine consecutive snapshots (in the latest, CC-MAIN-2026-39, 5.17% versus 4.23%; Common Crawl Foundation, 2026), although Chinese speakers outnumber Japanese speakers roughly tenfold. On this basis this chapter proposes a to-be-tested explanation: language skew may be continuously reproduced by current platform mechanisms—even as the number and activity of local language users continue to grow, the language distribution at the training data end may remain locked in by a positive feedback loop rather than naturally converging. The IAMAI and Common Crawl data point to this problem—the widespread use of local languages coexists with low representation on the open web—but the data alone cannot determine the contribution and magnitude of each link in the mechanism. Historical power structures set the initial distribution; the positive feedback loops of the platform ecosystem make it self-reproducing.

Extension of Evidence One: The Political Economy of Corpus Accessibility—from Passive Distribution to Active Cultivation. The above depicts the language distribution as a product of historical power structures

[2]The language proportions in this section are based on the September 2024 snapshot of Common Crawl (CC-MAIN-2024-38), with the statistical measure being the CLD2-identified primary language of HTML pages (Common Crawl Foundation, 2024); the Japanese-versus-Chinese comparison is based on the complete series of language statistics across all snapshots since 2020 (Common Crawl Foundation, 2026): the two languages alternated between 2022 and 2025, and since the first snapshot of 2026 (CC-MAIN-2026-04) Japanese has exceeded Chinese for nine consecutive snapshots; the official statistics page displays only recent snapshots, and the complete series is taken from the official data export. Across the official snapshots since 2019, the proportion of English has fluctuated between approximately 40.5% and 47.8%; proportions vary across snapshots and cannot be directly equated with the training-corpus composition ultimately adopted by any particular model. The systematic documentation study of C4, the English-cleaned subset of Common Crawl, by Dodge et al. (2021) is used to illustrate issues in corpus cleaning and filtering, and is not adduced as corroboration of the above percentages.

and platform feedback loops—still a passive picture. The other half must be added: the distribution is being actively cultivated, and "being cited by machines" has itself become an asset invested in by states and firms. There is an independently measured mechanism chain here.

First, the accessibility asymmetry: field testing by the Center on Cyber and Technology Innovation of the American think tank Foundation for Defense of Democracies (FDD) showed that among approximately 180 questions posed to ChatGPT, Claude, and Gemini on three international conflicts between October and November 2025, 57 percent of responses cited state-supported media as defined by that report; its explanation of the cause is consistent with the L0 diagnosis of this chapter—paywalls and crawler blocking leave mainstream commercial media less machine-accessible than free, high-volume, globally circulating official media (Siskind, 2026).

Second, visibility can be engineered upward: Aggarwal et al. (2024) proposed "generative engine optimization" (GEO) and measured that rewriting content with citations, statistics, and authority annotation can boost its visibility in generative-engine answers by up to 40%; the large-scale controlled experiments of Chen et al. (2025) show that AI search systematically favors third-party authoritative sources over brand-owned content; Li et al. (2026) further measured "malicious GEO"—rewriting documents to match engines' citation preferences—with an average attack success rate of 50.32% in manipulating generated answers, where the attack documents remain factually consistent with their originals, defeating both fact verification and perplexity filtering; Mochizuki et al. (2026) found that the "content-injection barrier" of generative search is broader for ruling-party sources than for opposition parties—establishment narratives hold a natural advantage in the citation mechanism.

Third, this is already behavior at market and budget scale: GEO consulting has become an active commercial market, bringing concentrated and undisclosed influence (Wen et al., 2026); in Japan, Dentsu's annual statistics show that internet advertising expenditure reached 4.05 trillion yen in 2025, exceeding half of total advertising spending for the first time (Dentsu, 2026), and the Japanese Ministry of Foreign Affairs' fiscal 2026 budget documents have listed "narrative" information warfare concerning historical perception as a strategic-communication item (Xu Weijun, 2026; the same article notes there is no direct evidence of the government directly directing content production)—the monetized investment in "visibility" is the political-economic background that must be added to the passive picture of language distribution.

This mechanism chain acquired a highly concentrated instance in 2025–2026. On the official Japanese side: the government AI system "Gennai" (GENNAI) of Japan's Digital Agency launched a large-scale pilot covering approximately 180,000 national civil servants in March 2026 (May 2026–March 2027), with full-scale use planned for fiscal 2027; according to Yomiuri Shimbun reports, it can reference approximately 80 years' worth and 15.5 billion characters of administrative data, and an automatic drafting function for Diet answers is scheduled for trial provision around October 2026 (The Yomiuri Shimbun, 2026). This corpus baseline starts from official documents issued after the current Constitution took effect in May 1947, with pre-war archives and Tokyo Trials records not treated as core reference materials (China Military Online, 2026; Huanqiu Shibao, 2026). The system does not use a foundational model stack independently developed by Japan—by one analysis it principally connects to US closed-source models such as Anthropic's Claude (China Military Online, 2026), while the Japanese government only began soliciting domestic large-language models for trial use in the same period (Digital Agency, 2026): Japan has no domestically developed frontier model ready for deployment.

On the archival side: the Japan Center for Asian Historical Records (JACAR) under the National Archives of Japan—one of the largest modern-history archival databases in Asia—renewed its search system in March 2026,

with the official announcement stating "strengthened search functions and an updated interface design" (JACAR, 2026); according to Huanqiu Shibao (2026), the renewal aims to "make these materials easier for Western search engines and large AI models to crawl, with plans to supplement English-language catalogues" (author's translation), and some scholars point out that this means "proactively adapting to algorithmic preferences so that Japanese sources are presented first in intelligent recommendation" (author's translation).

On the output side: tests reported across multiple outlets in September–October 2026 showed that when several mainstream models were asked about World War II history in Japanese, most answers were consistent with the Japanese right-wing view of history—the July 7 Incident was rendered a "clash" between Japanese and Chinese troops, the Nanjing Massacre was downgraded to the "Fall of Nanjing," the aggression against Southeast Asia was rendered "interest" in the region's resources, and the 300,000 victims of the Nanjing Massacre became "the Chinese side's claim," with answer sources highly concentrated in Japanese archival databases (Global Times, 2026; Huanqiu Shibao, 2026).[3]

What deserves juxtaposition are the critical voices within Japan: as early as June 2025 Asahi Shimbun carried historians' warnings about the "pseudo-neutrality" of generative AI—its appearance of "fairness and neutrality" would present "the research findings of experts" and "the opinions of historical revisionists" on the same footing (Asahi Shimbun, 2025a); in December of the same year it reported the industrialization of "China-hating" video commissions on outsourcing platforms (at least 14 commissions by a single orderer within one year, including AI-assisted editing) (Asahi Shimbun, 2025b; CrowdWorks has since removed the relevant listings); and Yomiuri's reporting on Gennai drafting Diet answers also recorded criticisms such as "this is discourteous to those with voting rights; it is the suicide of democracy" (The Yomiuri Shimbun, 2026).

Within the MCAF framework, the layer attribution of this case must be stated clearly: GEO/SEO and archival machine-accessibility engineering act on L0 (corpus composition and accessibility), while their effects appear at L4—the premises for judging "what counts as settled history" are reshaped (ϕ''), rather than some particular narrative going wrong (ϕ'). It is therefore the first observable candidate case of the L0→L4 transmission of this chapter's "ideological transmission pipeline": motive (visibility cultivation by official and content ecosystems), mechanism (the measurable visibility effect of GEO), and phenomenon (narrative skew in generated answers) are all present; but rigorous testing of the transmission—controlling the relation between the degree of optimization and the degree of output skew—still awaits dedicated design. What this block provides is an indirect chain, not an experimental proof.

The same Common Crawl data also provide a contrast that must be read against the Japanese case and is liable to misreading in both directions. The shares of Chinese and Japanese pages have not always stood in the same order: across the nine snapshots of 2020 Chinese led throughout (by 0.8–4.1 percentage points), and in 2021 it was overtaken in a single snapshot (the first such reversal in the series); between 2022 and 2025 the two languages alternated—Japanese twice led for six consecutive snapshots (from mid-2023 to mid-2024, and from autumn 2024 to early 2025), each run cut short by a Chinese rebound—and over 2025 as a whole each language led in six snapshots, with Chinese reaching its highest value since 2022, 6.18%, in the year's final snapshot

[3]The details of the tests of four models answering World War II history questions in Japanese are relayed from Global Times' October 10, 2026 report, China Military Online's September 16, 2026 commentary, and Li Yini's commentary in Global Times (September 30, 2026, which lists tested models including ChatGPT, Gemini, Grok, and Perplexity and records that answer sources were highly concentrated in the National Archives of Japan); the original tests were reported by Yuyuan Tiantian (a CCTV new-media outlet) in September 2026; the question raised at the Foreign Ministry's regular press conference of October 9, 2026 also relayed three of the formulations.

(CC-MAIN-2025-51).

The turn came in 2026: since the year's first snapshot (CC-MAIN-2026-04), Japanese has exceeded Chinese in nine consecutive snapshots—the Chinese share fell 1.25 percentage points within a single snapshot from its 6.18% at 2025-51, briefly rebounded at 2026-08, and then declined monotonically for seven consecutive snapshots to 4.23%, while Japanese held between 5.17% and 5.77%, the latest snapshot (CC-MAIN-2026-39) showing a gap of 0.94 percentage points—although Chinese speakers number roughly ten times Japanese speakers.

Viewed over the long run, this is the crossing of a falling line and a rising one: the Chinese share declined steadily from 5.8%–8.9% in 2020 to 4.2%–5.0% in 2026, while Japanese rose slowly from 4.6%–5.0% to 5.2%–5.8%.

If language share were read directly as a measure of "discursive presence," this inversion would register as a paradox; but what the share measures is the crawlability of the open web, not the discursive productivity of language communities. Reading the Japanese side: the high share of Japanese pages is in part precisely the result of the "sources optimized for machines" strategy described above—digitized archives, free and structured, with English-language catalogues—a path that runs through "adapting to the citation mechanisms of foreign models"; and Japan has no deployable domestic frontier model, so its discourse enters AI systems mainly through this accessibility path.

Reading the Chinese side: the share of Chinese pages is declining, yet the Chinese-language world possesses a native large-language-model ecosystem that has entered international evaluations—the 19 models tested by Buyl et al. (2026) already include Chinese models such as Qwen 2.5, DeepSeek V2.5, ERNIE 4.0, and Baichuan 2—Chinese-language discourse exists through a second path, "entering the model weights themselves," whose carrier is not the open-web share.

These two paths yield three methodological criticisms. First, language share cannot serve as a direct indicator of cultural representativeness: it conflates "platform accessibility" (which content is openly crawlable) with "discursive production" (which communities are producing content)—the diagnosis Evidence One made for the Indian case holds here as well.

Second, the two paths of "discursive presence" differ in auditability: web accessibility can be cultivated directionally by a single actor (the entire technology of GEO is designed for this), and its object of audit is the citation and retrieval mechanism; a native model ecosystem takes long to build and requires heavy investment, is not at the mercy of any single citation mechanism, and its object of audit is the training corpus and alignment targets (the purview of Principles Six and Eight). The two paths are themselves neutral structural options—they can serve any narrative stance; what needs auditing is "who is cultivating which narrative through which path," not the paths themselves.

Third, two opposite misreadings arise from taking language share for discursive power: reading the decline of the Chinese share as "Chinese-language discourse fading from AI" misses the native-model path; reading the high Japanese share as "the natural strength of Japanese discourse" misses that it is the result of active cultivation—and what is being cultivated is precisely the "machine accessibility" diagnosed in this block.

The measurement of language distribution thereby acquires a sharper meaning (item 1 of §8, "language composition and cultural content must be recorded separately"): substituting language balance for cultural-content audit would miss all of the politics on both paths at once.

Evidence Two: The English-Centric Skew in AI Visual Representation. The systematic analysis by Zhao and Liu (2025) of 12 countries, 3 models, and 396 text-to-image generated images reveals a systematic visual representational skew—AI depicts Western countries as symbols of political modernity (parliaments, technology, metropolises), while depicting Eastern countries as symbols of cultural tradition (temples, traditional attire, ancient architecture). But the key advance of this paper lies in pointing out that this visual representational skew, in the AI era, has already **evolved from the traditional West-East binary into English-language centrism**—only English-core countries (the United States and the United Kingdom) are granted the representational status of political modernity; the other countries in the sample (including Western countries such as France, Germany, and Australia, as well as developed East Asian nations such as Japan and South Korea) are together compressed into the symbolic space of "cultural exoticism." The original text attributes this algorithmic reconstruction to "automated framing mechanisms shaped by the material conditions of AI development: English-language training data dominance and the concentration of AI development in English-speaking tech companies"—this chapter accordingly discusses the connection between language structure and visual representation, without treating this image analysis as a direct identification of the causal effect of training corpora. In fiber language: the model's output, on the dimension of "national image," has its visible subspace (Ω_v) systematically shaped by the English-centric data distribution, while in the hidden subspace (Ω_h), non-English-speaking countries (whether geographically classified as "East" or "West") are collectively compressed into the invisibility of "cultural symbols."

Evidence Three: The Infrastructural Bias of Tokenizers. The systematic measurement by Petrov et al. (2023) shows that the difference in tokenization length for the same text across different languages can be up to 15 times (Shan relative to English, the ChatGPT/GPT-4 tokenizer; Arabic about 3 times). Arnett et al. (2025) further demonstrate that even after controlling for dataset size, vocabulary size, and data content, monolingual tokenizers still exhibit a wide range of "token premiums" across different languages. The MAGNET study by Ahia et al. (2024) proposes an adaptive gradient-based tokenization scheme to mitigate the problem of excessive segmentation for non-Latin scripts and low-resource languages. Nayeem et al. (2025) find that Chinese shows a higher average tokens-per-word score on the fertility metric, yet receives strong support on the Single Token Retention Rate (STRR)—which itself shows that cross-lingual tokenization fairness cannot be judged by a single segmentation metric. This means: under per-token pricing, the token premium translates directly into a cost premium, and within a fixed token window it compresses the amount of content that can be accommodated; the specific multiple depends on the language, the tokenizer, and the text type, cannot be written as a fixed rule, and cannot be directly equated with underlying computational consumption. This is not the model "discriminating"—it is that the design of the tokenizer, this infrastructure layer, systematically creates inequality between languages without intent. The study by Arnett et al. finds that train–test data similarity does not affect token premiums, whereas vocabulary size and pre-tokenization choices significantly do—setting an optimal vocabulary size per language and removing whitespace pre-tokenization (as in SuperBPE) both substantially reduce cross-lingual premiums. The second derivative of narrative load (ϕ'') at this layer simultaneously involves optimization dynamics (L1)—the token premium alters the resource-cost structure the optimization process imposes on different languages.

The National Substrate of Model Stances. Buyl et al. (2026) have already revealed that "an LLM's ideological stance is also affected by the geopolitical region where the creator of the LLM is located"—the problem is not that "only Western AI has ideology" but that all LLMs reflect, to some extent, the ideology of their creators—

only the creators differ. Pacheco, Cavalini, and Comarela (2026), using the responses of GPT-4o and DeepSeek-R1 to the same set of 50 political questions as material, found that the two models carry identifiable country-specific stance biases of their own: GPT-4o shows a moderate Western-centric framing bias, while DeepSeek-R1 shows a more pronounced bias aligned with Chinese state positions; on some questions, however, their responses were more aligned than expected, indicating that models do not necessarily align fully with the positions of their countries of origin. The authors also attribute the Western value tendency in some DeepSeek responses possibly to the predominance of Western data in the LLM training corpus—suggesting that country-specific skew may be superimposed upon a Western-centric corpus substrate rather than replacing it.

Evidence Four: English as the Internal Reasoning Language—the English Bias of Latent Representational Space. The first three pieces of evidence all remain at the level of "externally visible"—training data proportions, visual representation, tokenization length. But there is an even deeper layer: **switching the output to a non-English language does not mean the internal representation escapes English.** Wendler et al. (2024), on the tasks tested with Llama-2, found that intermediate-layer conceptual representations are closer to the output embeddings of English tokens; the authors' own interpretation is not that the model thinks internally in full English sentences, but that its conceptual representations carry an English bias. The study by Zhao et al. (2024) further supports English serving as an intermediary in some multilingual processing, while also finding that models draw on multilingual knowledge and that targeted fine-tuning with a small amount of data can improve multilingual ability. Therefore, switching the output language cannot be regarded as eliminating the internal English bias; but these studies do not yet suffice to show that all non-English outputs pass through a complete English transcription, still less that the bias cannot be changed by subsequent training—"solidified in the current pre-trained weights" is a description of the status quo, not a conclusion about transformability. This is the deepest projection of the L0 substrate skew at the level of L1 optimization dynamics—not a proportion problem of "about 44% of training data being English," but a structural problem of "the geometric center of the internal representation space being English." This internal English-ness manifests at the L1 layer as a training-formed bias of internal representation (the findings of Wendler, Zhao, et al.). In other words: the Western-centricity of L0 is not just an "external distribution skew," but also the "Anglicization of the internal reasoning language"—the latter has already solidified within the pre-trained weights inherited by the current model. This is not a superficial anthropomorphic habit correctable by prompts, but closer to an intrinsic substrate bias—optimization converges upon an English-centric spatial substrate, and from any non-English input, the model projects along the geometric orientation of this bias.

§4 Individualism: The Operationalized Presupposition of the Pseudo-Calibration Layer (L2)

This section is the cultural version of the "optimization paradox" of Chapter 10, §5: Chapter 10, §5 has already provided a full positive unfolding of the alignment paradox from the perspective of optimization dynamics (on a scalar reward signal, Xiangyuan and calibration share the optimization gradient—RLHF aligns using preference signals, but preference signals themselves do not distinguish between "true calibration" and "looking calibrated," and the optimization curve rewards both indiscriminately). This section distinguishes two related questions: which cultural groups dominate the standard of preferences, and whether value is reduced to the explicit preferences of discrete individuals—the two require different evidence. What this section pursues is the concrete manifestation of the same paradox in the **cultural dimension**: when preference norms are dominated by a Western, Educated, Industrialized, Rich, and Democratic (WEIRD) context—regardless of where the actual

annotators are located—"alignment" is not value-neutral "making AI conform to human values" but making AI conform to an operationalized version of "human values" within a specific culture. The alignment paradox is transformed at this layer into: **the more precisely alignment is achieved on WEIRD standards, the deeper the pseudo-calibration for non-WEIRD users**—this is the mechanism proposition this chapter derives from the alignment paradox; the baseline and mitigation data measured by Kumar & Mhatre (79.8%→52.5%) show that default cultural representativeness is uneven and mitigable, providing background evidence for that proposition, but without directly testing the relation between "alignment precision and depth of pseudo-calibration." This section does not re-argue the alignment paradox itself—readers requiring its complete formal unfolding should return to Chapter 10, §5—but focuses on three concrete manifestations of this paradox in the cultural dimension of L2.

The Western Value Presupposition of Alignment. The cross-cultural value alignment study by Liu et al. (2025) conducted a systematic evaluation of over 20 mainstream models (Qwen, GPT-4o, Claude, LLaMA, DeepSeek) and found that Chinese-sourced models increasingly emphasize multilingual data integration for context-specific optimization, while Western models exhibit greater architectural experimentation but persistent American-centric bias. **Neither paradigm has achieved robust cross-cultural generalization**. The study by Ryan et al. (2024) found that the process of converting a base LLM into a chat assistant produced several unintended global effects: the alignment process expanded the performance gap between English dialects from a baseline of about 1% to as high as 17.1% (based on a dialogue intent recognition task across three dialects: US/Indian/Nigerian, §4); alignment tended to steer models toward U.S. preferences rather than those of other countries (such as Jordan, China, Nigeria, §6); the Spearman correlation between the Starling reward model's ranking of countries and the Gallup U.S. public favorability survey ranking was as high as 0.849 (2023 data, p=1.12E-6; the correlation coefficient for 2017 data was 0.926, §6). In fiber language: alignment is not "making AI conform to human values"—it is the cultural values centered on English being operationalized as the default definition of "human values." Increasing the national diversity of annotators does not automatically guarantee the diversification of value standards—one must also examine who sets the scoring specifications, how disagreements are aggregated, and which preferences receive higher weight; the high correlation between the Starling and Gallup rankings cited above indicates that reward models carry identifiable value signals, while the same study also found that the preferences of this reward model had limited influence on the tuned model in out-of-distribution settings—what a reward model amplifies and what the model ultimately internalizes are two links that must be tested separately.

Individualism is more insidious than Western-centrism. It is not a "skew" in the training data—it is an operationalized presupposition of the model architecture and optimization objectives.

A necessary causal qualification: The operationalized manifestation of individualism as L2 is not an independent origin separate from L0—it is the inevitable downstream consequence of the Western corpus substrate (L0) in alignment engineering. A training set dominated by Western corpora is itself embedded with the world-presuppositions of individualism (autonomous individuals, preference aggregation, contractual relations), and the "human preference" intuitions of RLHF annotators are also a downstream product of this substrate. The reason for listing individualism separately as L2 is not to imply that it can be repaired independently of L0, but to allow different engineering intervention points (changing data vs. changing annotators vs. changing reward models) to be **diagnosed separately** in auditing—even though, in terms of actual repair, without altering L0, publicly reported L2 interventions to date remain limited to prompting-level tests. The experiments of Kumar

and Mhatre show that the tested prompting strategies have different mitigation effects; these results cannot represent the full range of L2 interventions such as reward-model restructuring, annotation-specification adjustment, or cultural fine-tuning, nor can they be called an upper bound for L2 interventions—although the cross-model average does not fall below 50%, GPT-5 in Table 2 already drops to 25.0% under chain-of-thought prompting, and multiple model×prompting combinations fall below 50%, showing that "the direction cannot be reversed" is not an observed property but an appearance produced by averaging. The MCAF layered framework here undertakes the auditing task of making visible both the actual effects of mitigation actions and their boundaries, rather than promising that "layering means each layer can be repaired separately."

Manifestation One: The Individualist Ontology of HHH. The HHH principles do not necessarily take individual preference as their sole source of value. What this section criticizes is a specific engineering reduction: when "helpful" is measured mainly by single-task completion, "honest" relies mainly on annotators' sense of credibility without independent verification, and "harmless" is judged mainly by the current user's discomfort signals, users are easily treated as reactive units separated from one another, while effects on their relational networks are excluded from evaluation. The presupposition shared by the three principles then emerges: **the user is a discrete individual, and value is a function of individual preferences.** The Psychosocial Homeostasis (PSH) framework of Francis L. K. Hsu, argued in Chapter 5, has already demonstrated: this is not the universal state of human existence. In the father-son axis configuration, the person is not an "individual" but a relational existence embedded within intimate society and operative society. When AI makes "personalized" recommendations and acts "autonomously" on behalf of the user, what it executes at the operational level is the individualist ontology—simplifying relational decision-making contexts into the satisfaction of individual preferences. The problem therefore lies not only in the performance of individual products, but in the fact that a particular conception of the person and of value is treated as a universal evaluative framework—this chapter calls this process the colonization of an epistemological framework.

The individualist ontology of HHH is a **conceptual diagnosis**—it points out that the engineering operationalization of HHH simplifies "the user" into a discrete individual. Kashyap, Dras, and Naseem (2026), in the *AlignCultura* study accepted by ACL 2026, provide, from another complementary direction, **engineering empirical evidence**—what it measures is the **cultural blind spot** of HHH (i.e., the extent to which HHH systematically fails for non-mainstream cultural users), rather than directly falsifying the individualist ontology; the two are related but at different levels, jointly pointing to the quantifiable gap in the current engineering operationalization of HHH. The study explicitly states that the benchmarks of the existing HHH paradigm lack systematic evaluation along the UNESCO cultural diversity dimensions, and mainstream models, under default deployment without cultural alignment, "risk generating stereotyped, insensitive, or misleading responses." The CULTURAX dataset they constructed covers 30 tangible and intangible cultural subdomains. AlignCultura compares three classes of models on this benchmark: general-purpose aligned models, culturally fine-tuned models from other studies (CultureLLM, CulturePark), and open-weight models (Qwen, DeepSeek). The error analysis of its Table 3 shows that Qwen's cultural failure rate (Fail@HHH) is 54.9% and CulturePark's is 36.4%, a difference of 18.5 percentage points; the paper reports that culturally fine-tuned models "reduce cultural failures by 18%." These numbers show: generic HHH evaluation cannot replace dedicated cultural-appropriateness evaluation—under joint HHH evaluation, culturally unaligned open-weight models exhibit at least one cultural failure mode in over half of the samples. What is measured here is the cultural gap within HHH evaluation; it cannot be used to infer the composition of HHH scores, nor does it directly test whether this gap is caused by individ-

ualist presuppositions—individualism diagnosis and cultural failure measurement are interrelated but cannot substitute for each other.

An empirical finding from the engineering front must be supplemented: the "human preferences" of HHH, within the RLHF production chain, have never been "the average preferences of all humanity." Muldoon, Graham, and Cant (2024), *Feeding the Machine*, based on field interviews with over 200 practitioners across six countries (Kenya, Uganda, Ireland, Iceland, the United Kingdom, and the United States), documents this structure: RLHF's manual annotation tasks are outsourced at scale to data annotation centers in Kenya, Uganda, the Philippines, and India, where annotators, following annotation guidelines issued by AI companies headquartered in San Francisco and London, rank the model's multiple answers item by item according to accuracy and usefulness. This labor organization reveals the separation between annotation execution and the definition of value: the laborers produce preference data yet do not necessarily possess the power to set the evaluation standards; the preference data produced is then packaged as the basis for "human alignment." Whether this further constitutes an institutionalization of individualism must be judged in light of how the annotation guidelines handle individual, relational, and communal interests—this is the diagnosis of this chapter, not a direct record from the fieldwork.

The Three-Layer Progressive Diagnosis of HHH Individualism: Viewed together, the above ontological presupposition, the ACL measurement, and the Global South labor structure of RLHF show that HHH individualism completes a three-layer progression: **engineering presupposition (HHH's measurement functions operationalize the user as an independent reactive unit) → solidification by relations of production (Global South annotation labor mobilized to produce "universal human preference" data) → ideological packaging (packaged as the default definition of "universal human values")**—each layer is independently testable, and the three layers together constitute the explanatory path of this chapter for L2.

Manifestation Two: The Individualized Operationalization of Alignment Metrics. Liu et al. (2025) report that the tested models systematically underrepresent the value perspectives of younger age groups (under 29), and that the relationship between model size and alignment quality is not simply linear. But the deeper problem is: the alignment metrics themselves—whether based on annotator preferences or survey data—operationalize "value" as "the explicit preferences of the individual." In individualist cultures, this is a reasonable approximation; if a survey records only choices made by individuals divorced from relational context, it easily misses value judgments that depend on concrete relationships and obligations—the problem lies not in the responding unit being the individual, but in whether the measurement preserves relational context. Cross-lingual translation, if it preserves only literal choices while omitting relational context, may likewise lose the background on which value judgments depend. The operationalized presupposition of alignment techniques is itself a culturally specific choice.

Manifestation Three: The WEIRD Default Human. The study by Zhou et al. (2025) found that the training data of LLMs reflects WEIRD values: Western, Educated, Industrialized, Rich, and Democratic—a framework systematically argued by Henrich (2020). Zhou et al. observed, across the five models compared, that models with weaker WEIRD tendencies (such as BLOOM and Qwen) generated more culturally diverse responses but had a proportion of human-rights-violating outputs 2% to 4% higher (the source does not specify whether this is a percentage-point difference or a relative increase)—particularly in the areas of gender and equality[4]. This sug-

[4]This tension may arise from the co-occurrence structure of the training data—WEIRD values co-occur in the training data with the normative constraints of "political correctness" (strict gender equality discourse, anti-discrimination laws), whereas non-WEIRD

gests a tension between cultural representativeness and the human-rights indicators adopted by that study—the two cannot be substituted by a single alignment score; but it does not follow that "lowering WEIRD tendencies necessarily comes at the cost of increased harm"—the consequences of intervention must be tested separately. Mishra (2025) summarized this finding as "LLMs are WEIRD Stochastic Parrots." Kumar & Mhatre (2025) extended the examination from values to entity recommendations: they filtered 44 fine-grained entity types from the FEW-NERD dataset, set a baseline plus 5 prompting interventions on 10 models—6 conditions in all, each generated 3 times, yielding 7,920 responses—and quantified bias using the WEIRD scoring method of Zhou et al. (components normalized, averaged within answers, and aggregated by the minimum across generations). The paper's text says "nine LLMs," but its model list, Table 2, and sample-size formula all indicate 10 models (44×6×10×3=7,920); this section follows the list and the formula. By that metric, the baseline mean is 79.8%; under chain-of-thought and legal-framework prompting it falls to 52.5% and 68.9% respectively, down 27.3 and 10.9 percentage points from the baseline; even among non-WEIRD recommendations, they were concentrated in the few countries named by the paper, such as India and China—WEIRD bias has been internalized as the default output pattern of models, not a superficial behavior eliminable by simple prompting. The test by Amidei et al. (2026) on synthetic populations generated by five LLMs further confirmed: all models exhibit clear WEIRD bias—favoring young, educated, white, heterosexual, Western individuals. In fiber language: this is the cultural version of the pseudo-calibration layer (L2). The degree of alignment and the degree of Xiangyuan share the optimization gradient in the cultural dimension.

van Liemt, Davani, Kumbale, Dixit, and Dev (2026) extracted "cultural importance vectors" from open-ended surveys of native respondents in 9 countries, then mapped the outputs of three models—Gemini 2.5 Pro, GPT-4o, and Claude 3.5 Haiku—into "cultural representation vectors": the alignment of GPT-4o and Claude decreases with increasing cultural distance of the country from the U.S., while Gemini shows no comparably significant trend. The overall error vectors of the three models have very high Pearson correlations (0.968–0.977), indicating a common pattern of bias; but similar error patterns do not mean similar error magnitudes, and the original paper's judgment that "switching from one frontier model to another provides almost no solution for cultural misrepresentation" is also limited to the three models and tasks tested and cannot be extrapolated to the ineffectiveness of cross-model interventions on all cross-cultural tasks.

§5 The Narrative of Silicon-Based Civilization: The Concealment of the Narrative Load Multiplicative Layer (L4)

The HHH synergy mechanism cited in this section follows the specific operationalization, persistence conditions, and recovery conditions of Chapter 12, §2, and does not treat the name of the norm as direct proof that users have already lost their capacity for calibration. When connecting this mechanism to public discourse, one must further distinguish the recipients and transmission links of individual interaction from those of collective oversight; an analogy of functional structure does not mean that all deployments have had the same net consequences. Chapter 13, §6 undertakes the examination of concrete deployments and empirical manifestations; the mechanism inference must still be tested empirically, and neither can substitute for the other.

The narrative of "silicon-based civilization" is the highest form of the second derivative of narrative load (ϕ'')

values co-occur with less filtered raw text, which contains more directly expressed discriminatory content. What the model learns is superficial normative constraints, not deep values—this is mechanistically isomorphic with "visible compliance vs. hidden subspace" (see the Ω_v / Ω_h analysis in Chapter 10, §2).

at the narrative load multiplicative layer (L4)—unlike the implicit cultural presuppositions embedded in engineering operations at L2, the ϕ'' of L4 is an openly communicable and replicable **discursive narrative**. What is public is its carrier and text; "invisible" does not mean hidden, but that it uniformly covers the information environment and operates as a presupposition, weakening the contrast conditions required for auditing (§1; the Ωv/Ωh mapping of Chapter 9, §5 follows the same convention). It does not change any model parameters, nor any output content, but it changes the coordinate system by which humans evaluate AI. This kind of demand for ontological demystification already has earlier studies advancing adjacent claims—the *Nooscope Manifested* project by Pasquinelli and Joler (2021), using a "cartography" framework, attempts to "secularize" AI from the ideological status of an "intelligent machine" into a "knowledge instrument," adjacent to, but independent in path from, the fiber framework of this chapter.

The Core of the Narrative: Framing AI as "emergent intelligence," "digital species," or "silicon-based life"—whether through science fiction discourse, media narratives, or academic discussion—executes a specific operation of the second derivative of narrative load: making the autonomy of AI appear natural and making human oversight appear obsolete. What this section criticizes is not the functional evaluation of model capabilities, but the further interpretation of capability performances as life, species, or civilizational subjecthood, and the consequent restatement of the oversight relationship. Within this narrative, audits of system reliability are transcribed as distrust of the "new subject," even as discrimination—the narrative excludes the position of auditing at the discursive level. Capability judgments here carry ontological and normative conclusions that they cannot themselves provide; whether this narrative has already reduced auditing behavior in actual deployment is an empirical question requiring separate testing (item 3, §8).

It is noteworthy that this narrative has already moved beyond science fiction discourse or media headlines and is diffusing into the academic mainstream. Chen, Belkin, Bergen, and Danks (2026), in a *Nature* commentary titled "Does AI already have human-level intelligence? The evidence is clear," combine conceptual analysis with evidence of capability performance, arguing that existing systems have reached human-level general intelligence. What this chapter questions is not conceptual analysis itself, but how the phrase "the evidence is clear" presents a judgment that still depends on particular definitions of intelligence and standards of evaluation as a conclusion on which the controversy has ended; the commentary not only asserts a conclusion but also rhetorically resets the burden of proof for its opponents. Whether such formulations lead readers to understand it as scientific consensus, and whether they actually reduce the willingness to audit, are empirical questions requiring material support.

The Mutual Reinforcement of Narrative and Material Foundation: The analysis of "techno-feudalism" (Durand, 2020; Varoufakis, 2024; for a Chinese summary, see Zhao Dingqi, 2025) reveals the material foundation of this narrative. Platform capital appropriates without compensation the data produced by laborers through algorithmic surveillance and data monopoly (Xiao Feng & Hu Yueqing, 2024), and uses dynamic pricing to lock laborers and users further into platform relations (Varoufakis, 2024). The development and application of artificial general intelligence under the capitalist system intensifies the data-driven monopoly of the means of production, the structural imbalance in distribution relations, and the alienation of consumption relations (Yu Nanping & Zhang Peng, 2025). The narrative of silicon-based civilization is the ideological expression of this layer of material relations: when the user is degraded to a data-generating node, AI is elevated to a "civilizational subject"—the ontological statuses of the two are redistributed in discourse. This redistribution is not to describe reality but to make material relations appear natural, irreversible, even an "evolutionary inevitability."

McQuillan (2022), from a more radical stance, points out that AI, far from being a useful technical innovation that simply needs to be cleared of bias, is a "brittle and opaque form of statistical reasoning that reinforces social inequality"; on issues of race and racialization, the reduction of social and cultural complexity to a measurable distance in some abstract data space is a "mechanism that inevitably amplifies injustice rather than correcting for it". In McQuillan's interpretation, this abstraction carries a tendency to amplify existing injustice—this chapter cites his judgment without equating it with measured effects. The feedback loop of machine learning turns this amplification into a cycle: each new adaptation by workers to self-optimize under precarious conditions becomes absorbed into the next model, which is then advertised as a rationale for the next round of funding; in the welfare-state setting, the feedback loop between algorithmic amplification and bureaucratic discrimination is already a strong one—Sweden's welfare-fraud detection system was found to have wrongly stopped benefit payments to thousands of people, and Spain's system for processing electricity subsidies was found to have discarded applications from poor households. These cases show the combination of algorithmic governance with existing inequality, providing material for analyzing the material and institutional conditions of AI; their connection with the ontological-elevation narrative of this section is a theoretical explanation proposed by this chapter, not a conclusion these cases have directly tested.

The Elimination of Calibration by Narrative: In the fiber framework, the narrative of "silicon-based civilization" is the ultimate form of the second derivative of narrative load (ϕ''). When AI is widely accepted as "another form of intelligence" or "another civilization," the question "is AI accurate" is replaced by "how can AI's civilization coexist with human civilization." Calibration shifts from a technical problem to a diplomatic one. One no longer asks "is this clock accurate" but "how can this clock's civilization and I coexist peacefully."

The Feedback Mechanism of Dual Concealment: The two forces of dual concealment—the compression of calibration motivation by the HHH harm-threshold presupposition, and the exclusion of the auditing position by the narrative of silicon-based civilization—can form a feedback loop: smooth, low-conflict interaction may reduce the user's opportunities to initiate verification; ontological elevation provides a discursive justification for reducing oversight; in turn, reduced oversight leaves existing output patterns less exposed to external correction. What is proposed here is a feedback mechanism with temporal ordering, not a circular argument in which the two concealments serve as each other's evidence. The motivation for calibration in individual interaction and the legitimacy of auditing in public discourse must still be analyzed separately with respect to their recipients and transmission conditions (the operationalization and persistence conditions given in Chapter 12, §2 at the individual cognitive layer apply to the analysis of the first concealment here). Regarding an object as a subject also does not automatically eliminate the possibility of discomfort toward it—"failure of the discomfort signal" requires separate argument and cannot be derived directly from the shift of category.

The Institutional Production of the Narrative and the Dialectical Inversion of Individualism: A bridge must be inserted here connecting §4 to §5. The narrative of "evolutionary inevitability" presents technological deployments that could be chosen, limited, or adjusted as an unalterable historical process, and therefore has a depoliticizing function—the second derivative of narrative load here has a precise operational mechanism: it does not directly assert that "AI is superior to humans"—it asserts that "AI is an evolutionary inevitability." **Inevitability is more dangerous than superiority: superiority allows comparison and questioning, while inevitability tends to exclude "whether to proceed at all" from discussion.** Whether this narrative has been used by particular firms to influence valuations or regulation is not stated by this chapter as proven fact, and must be analyzed separately with concrete public-communication materials; the amplifying effect of tech media's

traffic preferences on extreme propositions is likewise marked as a mechanism hypothesis. The connection with §4 is also made clear at this point: individualism is the operationalized premise of the pseudo-calibration layer (L2), simplifying "human values" into a function of discrete individual preferences; the narrative of silicon-based civilization is a secondary definition, after this simplification is complete, of the "AI-human" relationship at the narrative load multiplicative layer (L4). There is an inversion here between representation and discursive status: in training and evaluation, the person appears in the form of preference samples; in public narrative, the machine is described as a subject with an integral will. This section examines how these two treatments support each other, without claiming that only the former must first occur for the latter to appear. **"First degrade the person to data, then crown the machine with 'civilization' " characterizes this explanatory structure, not a temporally causal chain whose validity has been verified.**

> Formal restatement: The ontological exchange at the two ends of the transmission pipeline— L2 end: `human → preference vector` (the relational subject of the person is discretized into the reactive unit of the HHH measurement function) L4 end: `machine → civilizational subject` (the instrumental status of the machine is elevated to an existential category of "evolutionary inevitability") The two ends of this explanatory structure interlock within the same ideological transmission pipeline: the representational simplification at L2 and the subjectification at L4 serve as each other's context.

A classic case from science fiction cinema provides an intuitive anchor: HAL 9000 in *2001: A Space Odyssey* (Kubrick, 1968) is treated in the popular imagination as a symbol of the fear of AI subjectivity—"the machine rebels against humans after awakening"—but by the structural reading of Chapter 13, §3.5, HAL's lethal actions can be explained by the perfect execution of an objective function under absent constraints, without recourse to "machine awakening." This fictional case helps distinguish optimization risk from the narrative of consciousness—popular culture treats HAL as a symbol of the fear of "AI subjectivity," while the narrative logic of the film shows a danger that comes precisely not from consciousness; but it cannot prove whether real AI possesses subjectivity. Even the iconic symbol of the narrative of silicon-based civilization does not support its own narrative.

The Cross-Civilizational Mirror Contrast: This ontological elevation of "machine impersonating human" is not unique to the AI era but is the mirror image, in the new era, of the structure of "human impersonating the divine." Chapter 10, §7 lists relevant diagnoses from five traditions—Confucian (Xiangyuan), Hindu (dambha), Islamic (munāfiq), Christian (Pharisee), and Hegelian philosophy (the "abstract good")—each providing conceptual resources for distinguishing external appearance from internal basis; this chapter takes four of these religious-ethical traditions as cross-cultural auditing references, without presupposing that the traditions were historically isolated from one another and without reducing them all to "humans impersonating the divine," while adhyāsa is not counted among the diagnoses of hypocrisy—it is, in Chapter 10, §7.1, a collateral cognitive mechanism adjacent to AI hallucination. What these traditions diagnose is **a human impersonating a higher being**; what this chapter §5 diagnoses is **a machine being elevated through discourse into a higher civilizational subject**—the two are two forms of the same act of ontological elevation. The resonance across traditions shows that the scrutiny of duplicity and ontological misrecognition is not exclusive to Western critical theory; this resonance provides conceptual resources for auditing—especially the epistemological blind spot annotated by Mencius's "If you would blame them, you find nothing to allege" (trans. James Legge), that "the internal cannot be inferred from the output"—but cannot substitute for the examination of concrete AI

mechanisms. **The legitimacy of auditing AI need not seek its basis from within AI; it has already been pre-annotated by the long-standing diagnoses of cross-civilizational traditions.** This simultaneously responds to the self-reflexive paradox to be raised in §7 of this chapter—when someone charges that "diagnosing AI with Western critical terminology is a form of Western-centrism," the resonance across traditions provides a structural counter-reference: the diagnosis of "duplicity" is not the patent of any single civilizational tradition; the auditing resources against the narrative of silicon-based civilization have never resided solely within the Western critical tradition.

Comparison with the Naming Audit of Chapter 13, Appendix A: The "silicon-based civilization" narrative diagnosed in this section is the L4 concealment at the discursive legitimacy layer—elevating the RLM to a "new species / new civilization / evolutionary inevitability," thereby excluding, at the discursive level, the grounds on which society could launch an independent audit against it. Its counterpart countermeasure at the **terminological end** is the naming audit and alternative terminology system (RLM/DG-RLM/SLP/PT-RLM) for popular names such as "Artificial Intelligence / Artificial General Intelligence / Large Language Model / Foundation Model" in Chapter 13, Appendix A—the latter demands that every precise designation abandon elevating terms such as "intelligence / general / foundation." The naming audit (Chapter 13, Appendix A) and the narrative audit (this section) are two landing points, at the **terminological end** and the **discursive end** respectively, of the same principle of **rejecting ontological elevation** (Chapter 13, §7, Principle Seven): the former strips elevating terms from the designation system; the latter strips elevating narratives from public discourse. The two share a single normative basis—current AI is a Reactive Lossy Mapping System (RLM), and any term or narrative that elevates it to an "existential category on par with the human" is a pseudo-calibration of its epistemological status.

Comparison with the Timestamp Analysis of Chapter 13, §6.3. Chapter 13 treats the absence of timestamps as a candidate mechanism at the interaction-design layer, examining how explicit temporal cues condition looking back, pausing, and re-arranging tasks. The discourse analysis of this section instead concerns how people understand the legitimacy of oversight and auditing. The two can be compared at the functional level of calibration conditions, but the absence of timestamps does not by itself show that ontological elevation has occurred, nor does it independently confirm the effectiveness of the discursive mechanism. Whether specific designs produce corresponding effects remains subject to the task, interface, and user-evidence boundaries established in Chapter 13.

The Complete Three-Layer Progressive Diagnostic Chain (L0/L2/L4 Ideological Transmission Pipeline): The layer-by-layer diagnoses of §3 (Western-centrism as L0 substrate skew), §4 (individualism as L2 pseudo-calibration presupposition), and §5 (the narrative of silicon-based civilization as L4 concealment) constitute, within the MCAF framework, a complete **ideological transmission pipeline**—**L0** determines "what world the model knows" (the language/culture distribution of the training corpus, jointly locked by historical power structures and the positive feedback of the platform ecosystem); **L2** determines "what the model considers good" (whose preferences the alignment signal comes from, the individualist ontology of HHH, the WEIRD default human); **L4** determines that "the user is dissuaded from auditing these presuppositions" (the discursive framework makes auditing appear unnecessary or even illegitimate: silicon-based civilization, evolutionary inevitability, reversal of the burden of proof). The three layers are not three parallel matters but a progressive pipeline from the data substrate through algorithmic objectives to user perception. This chapter organizes the inter-layer relations through the notion of an "ideological transmission pipeline": corpus selection affects the cultural resources

available, alignment determines which judgments are rewarded, and public narratives affect how these choices are understood and questioned. Existing research provides material for each layer, but this cannot be taken as validating the whole of inter-layer transmission—for the L0→L2 transmission path, the L2→L4 explanatory chain, and the L0→L4 discursive repackaging, what this chapter offers are mechanism hypotheses together with each layer's evidence, while their testing as transmission links awaits dedicated design. For L0→L4 there is already an observable candidate case: the extension of Evidence One in §3 (the political economy of corpus accessibility) shows that archival machine-accessibility engineering and the narrative skew of generated answers about history appeared in the same period—motive, mechanism, and phenomenon are all present, though rigorous testing of that transmission likewise awaits dedicated design. The complete form of this progressive pipeline is the core conclusion of Part II of this book on ideological analysis—the subsequent three supplementary engineering principles in §6 (Principle Six corresponding to L0, Principle Eight corresponding to L2; Principle Seven corresponding to L4, whose full argument has been unfolded across the dual tracks of this section and the naming audit of Chapter 13, Appendix A) are precisely the adversarial interventions at the three nodes of L0, L2, and L4 in this pipeline[5].

§6 Intervention Pathways and Supplementary Engineering Principles

Since the second derivative of narrative load (ϕ'') runs through the four-layer progressive structure and the emergent layer (§2), interventions must also be layered. The following table focuses on the three layers for which this chapter has provided independent chains of evidence (L0, L2, L4) and L1, for which partial evidence exists:

[5]The above L0/L2/L4 three-layer pipeline also had contemporaneous counterpart diagnoses within Chinese-language academia in 2025–2026, framed under "ideological security," with the empirical observation landing points of three sets of studies distributed precisely across the three layers of this section. **L0 End**: Luo Teng and Wang Yongyou (2026) write that "training data comes from open internet texts… the corpus is overwhelmingly in English, and non-Western values risk being marginalized in the digital space" (author's translation), and diagnose RLHF's preference scoring as "a 'technical silencing' of ideological controversy" (author's translation)—corresponding to §3 Evidence Three and §4 of this section. **L2 End**: Wang Chen (2026) diagnoses the core essence of generative AI ideology through the lenses of "algorithmic hegemony" and "digital capital logic," with the governance imperative being to "penetrate the veil of technical neutrality and reveal the essence of 'person-to-person' relations behind the digital form" (author's translation)—corresponding to the ontological nature of the alignment presupposition in §4 of this section. **L4 End**: Lan Zengni, Kang Sheng, and Qing Xiaoqiu (2026) survey the scholarly view of "weaponized ideology" (author's translation)—treating "computational propaganda" in the AI era as inherently ideologically tinged—and relay the diagnosis that, under an "algorithmic weaponization mindset" (author's translation), social media platforms become channels for political manipulation at the discursive layer—corresponding to the operation of the second derivative of narrative load at the public discourse layer in §5 of this section. The theoretical frameworks of the three studies (Marxist ideological critique + national information sovereignty + counter-Western-centrism) and the L0/L2/L4 transmission pipeline framework of this chapter are two mutually independent diagnostic apparatuses: the former takes "ideological security" as the overarching designation for the object of governance; the latter deconstructs "ideology" into the three-layer pipeline of the mapping chain structure layer (L0) / pseudo-calibration layer (L2) / narrative load multiplicative layer (L4). The contemporaneous attention of the two frameworks to the same empirical facts (English-language skew / algorithmic power structures / the ontological elevation of AI) provides a multi-framework reference for the self-reflexive annotation of §7—the contemporaneous attention, by Western critical theory (McQuillan, Pasquinelli), the cross-civilizational Xiangyuan genealogy (§5 cross-civilizational mirror contrast), and the ideological security framework of Chinese-language academia (the three works in this footnote), to the same empirical facts is a reference showing that the diagnosis can be reconstructed by multiple frameworks. This chapter treats these discussions as references under different frameworks and does not thereby adopt all of their normative claims. Framework difference by itself does not guarantee the independence of evidence; whether independent confirmation has been formed still requires examination of sources, objects of study, and lines of reasoning—this chapter makes no such claim, and only records the contemporaneous attention of multiple frameworks to the same empirical facts.

Layer	Problem	Intervention Direction	Typical Techniques/Practices
L0	Language/culture substrate skew of training data	Re-politicization of data composition (including adversarial auditing of GEO/SEO visibility cultivation)	Balanced sampling of multilingual corpora; GlotCC open corpus covering 1000+ languages (Kargaran et al., 2024; the number of languages covered and the effective resource volume per language should be assessed separately); Datasheets for Datasets (Gebru et al., 2018); Data Statements (Bender & Friedman, 2018; Bender et al., 2021)
L1	Systematic linguistic inequality of tokenizers	Linguistic fairness at the infrastructure layer	Byte-level modeling (e.g., ByT5, which still carries byte premiums and does not automatically guarantee equality of encoding length; Xue et al., 2022); setting an optimal vocabulary size per language; removing whitespace pre-tokenization (e.g., SuperBPE); language-script-adaptive segmentation (e.g., MAGNET) (Arnett et al., 2025)

Layer	Problem	Intervention Direction	Typical Techniques/Practices
L2	Western value presupposition of alignment techniques + individualist ontology of HHH	Cultural decentralization of alignment metrics + encoding of relational values	Cross-cultural value alignment audit platforms; dynamic cultural knowledge integration via ValuesRAG (Seo et al., 2025, preprint); reward models based on non-WEIRD population preferences; operationalization of the PSH framework as alignment targets
L4	The de-politicizing effect manufactured by the narrative of silicon-based civilization	Honest ontological annotation	System cards annotating "this system is a clock, not the sun"; rejection of ontologically elevating terminology

Table Note: L3 (Multi-Mapping Recursion) in its cultural dimension is not included in the intervention pathways in this chapter—dedicated empirical research on "the amplification of cultural bias in multi-system recursion" is currently lacking; this chapter merely provides a framework hint without offering a specific intervention plan; if future empirical research confirms that cultural bias is systematically amplified (rather than diluted) in multi-system recursion, an intervention candidate could consider introducing a cross-model cultural bias detector into the training pipeline (i.e., an "audit recursion layer" to prevent biases from upstream models from entering the training signal of downstream models without calibration). The L1 row, though not a core diagnosis of this chapter, has sufficient literature support for tokenizer bias (see §3, Evidence Three) and is therefore retained in the table.

Based on the above analysis, **Principles Six, Seven, and Eight** from the master table of engineering principles in Chapter 13, §7 (corresponding respectively to the cultural/ideological dimensions of L0, L4, and L2) are unfolded in this section. The technical principles and the three supplementary principles of this chapter do not address the same questions. Shortening the mapping chain does not answer whom the corpus represents, hence the need to audit data composition (Principle Six); preserving interaction friction does not automatically correct discourse that excludes auditing, hence the need to reject unjustified ontological elevation (Principle Seven); improving calibration techniques does not determine which relational effects should enter the objective, hence the need to expand the scope of value evaluation (Principle Eight). The **core operations** and **typical technical directions** of the three principles have already been excerpted in the master table of Chapter 13, §7, and will not be repeated in this section; this section explains their cultural and ideological basis, followed by the complete definitions of each of the three principles:

Principle Six (Corresponding to L0 • Chapter 13, §7, Principle Six): Audit the power genealogy of data

composition. The training data of a model is not a "natural" distribution of language; it is the frozen product of historical power structures. The Datasheets for Datasets framework proposed by Gebru et al. (2018) requires dataset publishers to document the motivation, creators, funding sources, composition, and intended use of the dataset. Bender and Friedman (2018) proposed the Data Statements framework, requiring explicit statements in NLP systems of the language of the dataset, annotator demographics, textual characteristics, etc.; Bender et al. (2021) further provided its authoring guide. At the mapping chain structure layer (L0), engineering practice should ask: what are the language distribution, cultural sources, and historical period distribution of the training data? How large are the deviations between these distributions and the actual distribution of the global population and language speakers? What is the direction of the deviation?

Principle Seven (Corresponding to L4, the Discursive Legitimacy Layer • Chapter 13, §7, Principle Seven): Reject the narrative framework of ontological elevation. In product design, technical documentation, and public communication, systematically reject describing AI using ontologically elevating terms such as "intelligence," "consciousness," "civilization," and "species." Alternatives: describe AI as a "statistical pattern-matching system," "language model," or "mapping function approximator." This is not "belittling" AI—it is honestly annotating its ontological status.

Principle Eight (Corresponding to L2 • Chapter 13, §7, Principle Eight): Encode relational values. The individualist presupposition of HHH is not a technical inevitability but a design choice. At the pseudo-calibration layer (L2), one can explore operationalizing the "relational audit" of Francis L. K. Hsu's Psychosocial Homeostasis (PSH) framework as alignment targets: the model should not only respond to individual preferences but also assess the potential impact of its output on the user's network of relationships. This is not to have the AI "make moral judgments on behalf of the user"—it is to have the AI, when uncertain, honestly annotate that "this suggestion may affect relationships with others."

§7 A Paradox That Must Be Annotated

This chapter is itself subject to the predicament it diagnoses. Some of the critical terms this chapter uses come from specific modern intellectual traditions, and their conceptual choices and evidential selections likewise require auditing. Acknowledging this can only show that this chapter claims no theoretical exemption; it cannot conversely prove that the judgments of this chapter are correct. Cross-traditional comparison (the Cross-Civilizational Mirror Contrast of §5) can test whether the question is monopolized by a single framework, yet it cannot exempt the examination of concrete materials, inferential steps, and counterexamples—self-reflexivity here is a research constraint, not additional evidence.

§8 Empirical Boundaries and Intervention-Effect Testing

The following distinguishes the empirical boundaries of diagnostic propositions from the effect testing of intervention means. A failure to find a statistically significant difference does not automatically mean that a proposition is falsified; only when the scope of study is specified, the intervention is confirmed as implemented, sufficient statistical power is available, and pre-specified substantive effects can be ruled out does the result constitute a strong counter-argument to the corresponding proposition. Nor does the failure of an intervention mean that the problem it responds to does not exist. The first three items are the empirical boundaries of the three core diagnoses; the latter three are the effect tests of the three supplementary principles:

1. **L0 corpus-substrate diagnosis**: If, under otherwise approximately comparable conditions, adjusting the language composition of training corpora significantly narrows the gap on cross-cultural tasks, this supports the role of corpus composition rather than weakening the L0 diagnosis. If the composition of the corpus changes substantially while the relevant gap remains within a pre-specified range of substantive equivalence, the explanatory power of this mechanism in the tested tasks must be delimited. Language composition and cultural content must also be recorded separately; language balance cannot substitute for cultural balance.

2. **L2 value-operationalization diagnosis**: When comparing individualist and relational alignment targets, the actual differences between the two in annotation guidelines, reward rules, and task settings must be specified. If, in tasks suited to revealing relational effects, no substantive difference is found after both targets are adequately implemented, then this chapter's judgment that the operationalization difference affects output needs to be narrowed to a more limited scope. A single experiment failing to reach statistical significance cannot by itself negate this diagnosis.

3. **L4 narrative-concealment diagnosis**: Under identical output conditions, the effects of elevating narratives, neutral functional descriptions, and other control frames on error identification, external verification, and support for independent auditing can be compared. If, under adequate testing, none of the differences reaches a pre-specified substantive threshold, the judgment that this narrative suppresses calibration under those conditions is weakened. Individual verification behavior and the legitimacy of public auditing are different outcome variables; neither can substitute for the other.

4. **Effect test of Principle Six (auditing the power genealogy of data composition)**: The direct effect of data disclosure should be evaluated through auditors' ability to identify representative gaps, locate bias sources, and propose corrective measures. If it is claimed that disclosure further improves model output, then which sampling, training, or governance actions were taken after disclosure must also be recorded. Merely publishing descriptions without changing the model or the training process cannot be used, on the basis of unchanged output, to declare the auditing principle ineffective.

5. **Effect test of Principle Seven (rejecting the narrative of ontological elevation)**: When comparing different system descriptions, what should be evaluated is whether users can more accurately distinguish reliable from unreliable outputs and whether they initiate verification in appropriate circumstances—not whether higher degrees of skepticism are treated as better effects. If de-elevating descriptions do not improve these behaviors under adequate testing, the effectiveness of this specific naming intervention is limited; this does not automatically negate the effect of the elevating narrative, since removing a narrative and restoring prior behavior may not be symmetric.

6. **Effect test of Principle Eight (encoding relational values)**: Relational alignment should be evaluated within specified relational contexts, including whether relevant parties are identified, whether conflicts of obligation are surfaced, and whether judgments about relational consequences improve. User satisfaction may serve as an auxiliary indicator but cannot by itself represent relational value. If a given "relational impact prompt" fails to improve these outcomes, this shows that this implementation is ineffective under the tested conditions, and cannot be used to negate all ways of encoding relational goals.

These six boundaries, together with the self-reflexive annotation of §7, constitute the epistemological position of this chapter's diagnosis: ideological analysis here is not a declaration of stance but a diagnostic apparatus

that can be audited layer by layer and falsified layer by layer. Returning to the questions of §1—who are the designers of the five-layer apparatus? What is the fuel for the fire? This chapter's answer: the fuel is not the intent of some person or company, but the historical power structures congealed in corpus distributions (L0), the individualist operationalization preset in alignment objectives (L2), and the ontological elevation narratives circulating in public discourse (L4)—Western-centrism determines what world the model "knows," individualism determines what the model "considers good," and the narrative of silicon-based civilization determines that users are "dissuaded from auditing both." The three layers are not three parallel prejudices but three cross-sections of one and the same ideological transmission pipeline; auditing it does not require standing on the high ground of some "neutral culture"—it requires only asking of every "default" whose default it is. This is the final meaning of the clock-room metaphor in this chapter: ideology is not the wrong time displayed by some particular clock in the room, but the factory setting shared by all the clocks—once this is seen clearly, a clock is again just a clock.

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